

2025 Fall Course Syllabus - Mathematics 3031.001

Course Title:

Probability Theory I

Course Credits:

3

Course Mode:

In person.

Course Days and Time:

MWF 11:00-11:50AM.

Course Room:

Wachman 00015

Course Instructor:

Wei-Shih Yang

Instructor Email:

yang@temple.edu

Instructor Office:

Wachman 534

Instructor Phone:

215-983-3816

Office Hours:

MWF 2:00-3:00PM.

Course Materials:

Introduction to Probability (Cambridge University Press; 1 edition, November 2, 2017) by Anderson, Seppalainen, and Valko, ISBN-13: 978-1108415859.

Course grading scheme:

Homework: 3%; Quizzes: 12%; Midterm 1: 25%; Midterm 2: 25%; Final: 35%. NO MAKE UP QUIZ or ADDITIONAL QUIZ can be given under any circumstances (including illness). However, lowest quiz score will be dropped. In other words, you may miss at most one quiz without hurting your grade. Correspondence between the numerical and letter grades: 93-100 A, 90-92 A-, 87-89 B+, 83-86 B, 80-82 B-, 77-79 C+, 73-76 C, 68-72 C-, 65-67 D+, 55-64 D, 50-54 D-, 0-49 F.

Course prerequisites:

Calculus II.

Course goals:

Students will learn basics of probability theory and develop technical skills to perform calculations.

Topics covered:

Discrete and continuous random variables, independence and conditional probability, probability distributions for one or several random variables, expected value and variance, moment generating functions, law of large numbers and central limit theorem, Poisson processes, conditional expectation.

Exam dates:

To be announced.

Attendance policy:

Attendance is required. More than 6 absences in total may affect the student's grade. If you miss a class due to illness, please provide a doctor's note to the instructor. Your absence in that class will not be counted towards the 6 absences.

Technology Specifications for this Course:

A calculator will be needed.

Disability Statement:

Any student who has a need for accommodations based on the impact of a documented disability or medical condition should contact Disability Resources and Services (DRS) in Howard Gittis Student Center South, Rm 420 (drs@temple.edu; 215-204-1280) to request accommodations and learn more about the resources available to you. If you have a DRS accommodation letter to share with me, or you would like to discuss your accommodations, please contact me as soon as practical. I will work with you and with DRS to coordinate reasonable accommodations for all students with

documented disabilities. All discussions related to your accommodations will be confidential.

Academic Freedom:

Freedom to teach and freedom to learn are inseparable facets of academic freedom. The University has adopted a policy on Student and Faculty Academic Rights and Responsibilities (Policy # 03.70.02) which can be accessed [here](#) (opens in new tab/window).

Attendance and Your Health:

To achieve course learning goals, students must attend and participate in classes, according to your instructors' requirements. However, if you feel unwell or if you are under quarantine or in isolation because you have been exposed to the virus or tested positive for it, you should not come to campus or attend in-person classes or activities. It is the student's responsibility to contact their instructors to create a plan for participation and engagement in the course as soon as they are able to do so, and to make a plan to complete all assignments in a timely fashion, when illness delays their completion.

Add/Drop Policy:

Students will be charged for a course unless dropped by the Drop/Add deadline date. Check the [University calendar](#) (opens in new tab/window) for exact dates.

During the Drop/Add period, students may drop a course with no record of the class appearing on their transcript. Students are not financially responsible for any courses dropped during this period. In the following weeks prior to or on the withdrawal date students may withdraw from a course with the grade of "W" appearing on their transcript. After the withdrawal date students may not withdraw from courses. Check the [University Calendar](#) (opens in new tab/window) for exact dates. [See the full policy by clicking here](#) (opens in new tab/window).

Incomplete Policy:

The grade "I" (an "incomplete") is only given if students cannot complete the course work due to circumstances beyond their control. It is necessary for the student to have completed the majority of the course work with a passing average and to sign an incomplete contract which clearly states what is left for the student to do and the deadline by which the work must be completed. The incomplete contract must also include a default grade that will be used in case the "I" grade is not resolved by the agreed deadline. [See the full policy by clicking here](#) (opens in new tab/window).

Exam Security Policy:

We have a zero tolerance policy towards cheating. Students caught cheating on a problem in a test (receiving outside help or using unauthorized resources) will receive a score of 0 for the entire test. This is consistent with the Temple University Academic Honor Code (see <https://secretary.temple.edu/sites/secretary/files/policies/03.70.12.pdf>) that states: "Every member of the university community is responsible for upholding the highest standards of honesty at all times. Students, as members of the community, are responsible for adhering to the principles of academic honesty and integrity". More serious penalties are possible and those caught cheating will be reported to the office of student conduct.